PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Chapter OL-T38 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

7 classified items • 34 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T38 • Expertise Q16+ • 7 items • 34 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Differentiation

Unit 2 • Expertise • 34 marks • 7 item(s)

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

(a) Find $\frac{dy}{dx}$

$\frac{dy}{dx} =$ ....
(2)

(b) Find the coordinates of the turning points on the graph with equation $y = 4x^3 + 5x^2 + 2x$
Show clear algebraic working.

....
(4)

(Total for Question 17 is 6 marks)

WORKING SPACE

4MA1-2H Q18 whole **4 marks**

18 A particle is moving along a straight line that passes through the fixed point O
The displacement, s metres, of the particle from O at time t seconds is given by

$$s = 2t^3 - 5t^2 + 6t - 5$$

Find the value of t when the acceleration of the particle is 5 m/s²

$t =$ ....

WORKING SPACE

4MA1-1H Q20 M01 **4 marks**

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.
Show clear algebraic working.

....

WORKING SPACE

4MA1-1H Q24 **6 marks**

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates (2, –6) lies on **C**
The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3
Show clear algebraic working.

$y =$ ....

WORKING SPACE

4MA1-2H Q24 **5 marks**

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geqslant 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is a m/s^2

Find the value of a

a = ...

WORKING SPACE

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x, in terms of p and m, for which the gradient of **C** is negative.

....

(Total for Question 23 is 4 marks)

YOUR WORKING

20 The radius of a right circular cylinder is x cm.

The height of the cylinder is $\left(\dfrac{800}{\pi x} - x\right)$ cm.

The volume of the cylinder is $V\,\text{cm}^3$

Find the maximum value of V
Give your answer correct to the nearest whole number.

YOUR WORKING

(Total for Question 20 is 5 marks)

YOUR WORKING